Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT305	Jimin XIAO	Intelligent Science	3209

Final Exam 2023/2024

BACHELOR DEGREE – Year 4

Machine Learning

TIME ALLOWED: 3 Hours 0 Minutes

INSTRUCTIONS TO CANDIDATES

1、 **Total marks available are 100.**

2、 **Answer all questions.**

3、 **The number in the column on the right indicates the approximate marks for each section.**

4、 **In general, it is particularly important to give reasons for your answer. Only partial marks will be awarded for correct answers with inadequate reasons.**

5、 **Answer should be written in the answer booklet(s) provided.**

6、 **The university approved calculator - Casio FS82ES/83ES can be used.**

7、 **Only solutions written in English will be accepted.**

Answer All Questions

1. Pooling units take n values $x_i,\ i \in [1, n]$ and compute a scalar output whose value is invariant to permutations of the inputs.

(1) The Lp-pooling module takes positive inputs and computes $y = (\sum_i x_i^p)^{\frac{1}{p}}$, assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x_i' = \frac{\partial L}{\partial x_i}$? PPT有

(2) The log-average module computes $y = \frac{1}{\beta} ln(\frac{1}{n} \sum_i exp(\beta x_i))$, assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x_i' = \frac{\partial L}{\partial x_i}$?

Total

16 (each 8)

2. Recall two linear classification methods we considered:

① 实数值 t∈{0,1}
model2 → prob
② 均方差范围差异大
0 → [0,1]
③ 非线性表达力

Model 1:

$$y = \mathbf{w}^T \mathbf{x} + b$$

$$\mathcal{L}_{SE}(y, t) = \frac{1}{2}(y - t)^2$$

Model 2:

$$z = \mathbf{w}^T \mathbf{x} + b$$

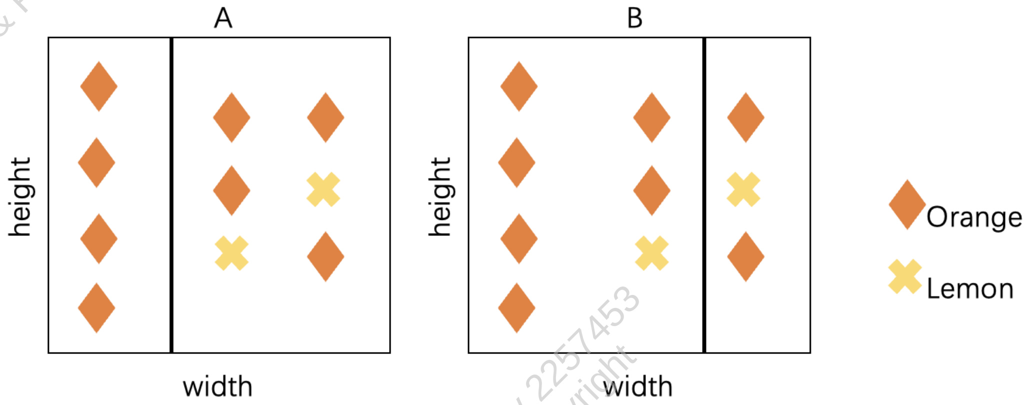
$$y = \sigma(z)$$

$$\mathcal{L}_{SE}(y, t) = \frac{1}{2}(y - t)^2$$

Here, σ denotes the logistic function (sigmoid function), and the target label t take values in $\{0, 1\}$. Briefly explain our reason for preferring Model 2 to Model 1.

Total
9

3. We are constructing a decision tree. Considering the following data, we would like to split them on width for one level. There are 2 options to split them as in the following figure. Please select the better one from the perspective of information gain.



Total 15

4. Convolution operation is defined as follows:

$$f[x,y] * g[x,y] = \sum_{n_1=-\infty}^{\infty} \sum_{n_2=-\infty}^{\infty} f[n_1,n_2] \cdot g[x-n_1, y-n_2]$$

The image patch is defined as follows

$$f = \begin{pmatrix} 1 & 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 3 \\ 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 3 & 1 & 1 \end{pmatrix}$$

The convolution kernel is defined as

$$g = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

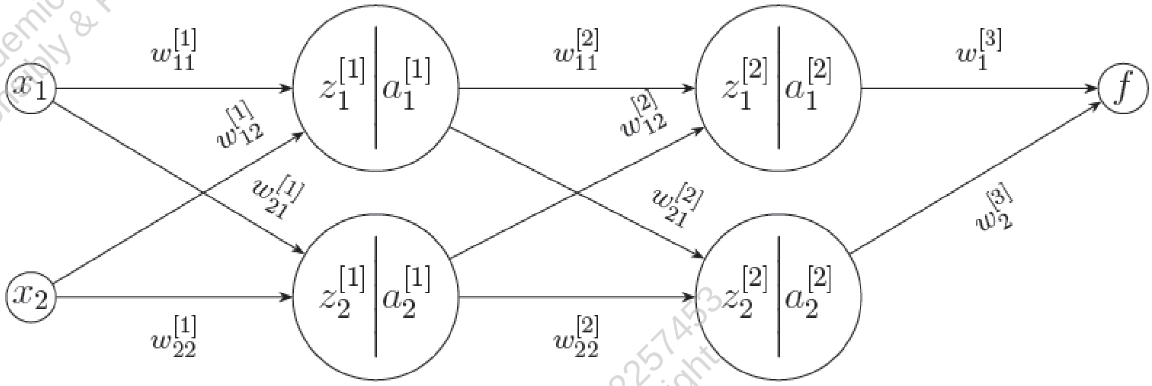
Please evaluate the convolution results for the 2 following cases:

1. Stride is equal to 1, and padding is NOT used. **(10)**
2. Stride is equal to 2, and zero padding is used. **(10)**

$(5+2\times 1-3)/2+1=3.$

Total 20

5. Consider this three-layer network, where the activation function $\sigma()$ is the sigmoid function, and $\sigma'(x) = \sigma(x)(1-\sigma(x))$.



$$Z^{[1]} = \begin{bmatrix} z_1^{[1]} \\ z_2^{[1]} \end{bmatrix} = \begin{bmatrix} w_{11}^{[1]} & w_{12}^{[1]} \\ w_{21}^{[1]} & w_{22}^{[1]} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad , \quad A^{[1]} = \begin{bmatrix} a_1^{[1]} \\ a_2^{[1]} \end{bmatrix} = \begin{bmatrix} \sigma(z_1^{[1]}) \\ \sigma(z_2^{[1]}) \end{bmatrix}$$

$$Z^{[2]} = \begin{bmatrix} z_1^{[2]} \\ z_2^{[2]} \end{bmatrix} = \begin{bmatrix} w_{11}^{[2]} & w_{12}^{[2]} \\ w_{21}^{[2]} & w_{22}^{[2]} \end{bmatrix} \begin{bmatrix} a_1^{[1]} \\ a_2^{[1]} \end{bmatrix} \quad , \quad A^{[2]} = \begin{bmatrix} a_1^{[2]} \\ a_2^{[2]} \end{bmatrix} = \begin{bmatrix} \sigma(z_1^{[2]}) \\ \sigma(z_2^{[2]}) \end{bmatrix}$$

Given that $f = w_1^{[3]} a_1^{[2]} + w_2^{[3]} a_2^{[2]}$, compute :

a) $\delta_1 = \frac{\partial f(x)}{\partial z_1^{[2]}}$ **(5)**

b) $\delta_2 = \frac{\partial f(x)}{\partial z_1^{[1]}}$ **(5)**

c) $\delta_3 = \frac{\partial f(x)}{\partial z_2^{[1]}}$ **(5)**

d) $\delta_4 = \frac{\partial f(x)}{\partial w_{11}^{[1]}}$ **(5)**

Total 20

6. We will be working with clustering by GMM for data $\mathcal{D}$.

Assume a datapoint $\mathbf{x}$ is generated as follows:

- Choose a cluster z from $\{1, . . . , K\}$ such that $p(z = k) = \pi_k$
- Given z, sample $\mathbf{x}$ from a Gaussian distribution $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_z, \mathbf{I})$

(1) What is $p(x)$?

(2) What is the log-like loss function to be maximized?

(3) If we want to use Expectation-Maximization algorithm for optimization, derive the E-step for this model.

Draw a rectangle around the formula that you derived.

(4) Derive the M-step for this model. **Draw a rectangle around the formula that you derived.**

Total
20 (each 5)

The end of the paper